

Outcomes

SAS dataset name: Outcomes

General description:

This dataset contains data on outcomes or endpoints for knee replacements, hip replacements, deaths, and x-ray outcomes. Participants are only included in this dataset if they have had an outcome as defined below.

IMPORTANT: Additional and revised outcome/endpoint data will be included in this dataset, and analysts are encouraged to ensure they are using the most up-to-date version of the dataset. Analyses ran using an older version of this dataset should be re-run using the most recent version of this dataset.

Variable naming conventions:

Variables in this dataset are related to endpoints or outcomes and as such, may not be associated with a specific visit and have the visit prefix “V99.” For a complete list of variable prefixes, please see the “*Visit Prefix Definitions*” found in the Outcomes data section. Endpoint-related variables in this dataset all have names which begin V99E (“E” for endpoint).

There are 3 sets of variables, one for knee replacements, one for hip replacements, and one for x-ray (radiographic) outcomes, for which there are both right knee and left knee versions of the variables. There is also a set of variables related to death which have names prefixed “V99ED...” In addition, there is a variable for the most recent OAI contact (see Table 7). The following table summarizes how the prefix of the variable name describes which set of variables it belongs to:

Table 1. Variable name prefixes and associated endpoint/outcome

Variable Name Prefix	Endpoint	Variables include data on
V99ERK....	Right Knee Replacement	Date (or visit) of replacement, whether adjudicated or seen on x-ray, days from enrollment to replacement, etc. (see Tables 3 & 4)
V99ELK....	Left Knee Replacement	
V99ERH....	Right Hip Replacement	
V99ELH....	Left Hip Replacement	
V99ERX..../ V99ERK..../ V99ERN..../ V99ERJ....	Right Knee X-ray	Prevalence/incidence of radiographic OA, medial and/or lateral progression of joint space narrowing, etc. (see Table 6)
V99ELX..../ V99ELK..../ V99ELN..../ V99ELJ....	Left Knee X-ray	
V99ED....	Death	
		Date of death, and whether reported or adjudicated from medial records, etc. (see Table 5)

The remaining letters in the variable name relate to the meaning of the variable. See Tables 3-7 below for a complete list of variables names, descriptions, values, and value labels.

Hip and knee replacements:

Participants with hip or knee replacements are included in the Outcomes dataset if one of the following three criteria were met:

- (1) The replacement was centrally adjudicated: knee and hip replacements that occur after a participant enrolled in the OAI study, the medical records forwarded by the OAI clinical center that document the event were reviewed by two adjudicators independently. If the adjudicators do not agree on an aspect of the event, a physician adjudicator reviews the medical records and/or x-ray and resolves the discrepancies.
- (2) The replacement was seen on an OAI study x-ray: all knee and pelvis x-rays were read for the presence of a knee or hip replacement.
- (3) The replacement was self-reported: participants that self-reported a replacement that has not gone through the adjudication process due to a lack of medical records and/or x-rays are also included in the Outcomes dataset.

Description of replacement variables:

Knee and hip replacement status during follow-up. ExxRPCF is the key variable for identifying knee and/or hip replacements that occurred after a participant enrolled in the study. For participants who have reported a knee or hip replacement during their follow-up interviews, this variable gives information on where participants are in the adjudication process and has four possible values:

- ‘0: Replacement adjudicated, failed to confirm:’ this value is assigned when a reported replacement was determined through the adjudication process to not be a true replacement (e.g., surgery was for a different procedure). Participants for whom it was determined through the adjudication process that a reported replacement did not occur are generally not included in the Outcomes dataset unless they are in that dataset for another reason (e.g., replacement on contralateral side or other joint).
- ‘1: Self-reported replacement:’ this value is assigned when a knee or hip replacement is reported by the participant, medical records cannot be obtained, and there are no x-rays for the visit(s) following the reported replacement from which to confirm the replacement.
- ‘2: Replacement pending adjudication:’ this value is assigned if a replacement has been reported and the adjudication process has not yet been completed. This value may change in subsequent releases of the Outcomes dataset as medical record documentation is received and/or x-rays are acquired for these participants.
- ‘3: Replacement adjudicated, confirmed:’ this value is assigned if a replacement has been confirmed through the adjudication process from medical records and/or post-operative x-rays.
- ‘.A:’ this SAS special missing value code denotes that no replacement has been reported for that joint and side.

To select all participants with replacements that occurred after enrollment and confirmed through the adjudication process use the ExxRPCF= 3 for both sides. For instance, to select all participants with confirmed **knee** replacements use **ERKRPCF= 3** and **ELKRPCF=3** (right and left, respectively); to select all participants with **hip** replacements use **ERHRPCF=3** (right hip) and **ELHRPCF=3** (left hip). To select

participants with a confirmed replacement or those who have a self-reported replacement but whose replacement has not been confirmed through adjudication, use ExxRPCF=1, 2 or 3.

Baseline knee or hip replacements. If a replacement was seen on the *baseline* OAI x-ray, this is noted in the variable ExxBLRP. To select all participants with baseline replacements use ExxBLRP=1 for both sides for the joint of interest.

Knee or hip replacement seen on follow-up OAI x-ray. If a replacement has been seen on a *follow-up* OAI x-ray, this is noted in the variable ExxRPSN regardless of whether a replacement has been reported by the participant.

- ‘0: No replacement is seen on any available follow-up x-ray:’ this value is assigned when a knee or hip replacement has not been seen on a follow-up x-ray. In cases where the value of ExxRPSN=‘0’ and the confirmation/adjudication status variable ExxRPCF confirms a knee or hip replacement, it is likely that the last OAI x-rays acquired and reviewed for evidence of the joint replacement were acquired before the replacement occurred. The value for ExxRPSN may change in the future if additional x-rays are acquired for the participant.
- ‘1: Yes, replacement is seen on one or more available follow-up x-rays:’ this value is assigned if a replacement has been seen on a follow-up x-ray. When the confirmation/adjudication status variable ExxRPCF indicates the adjudication is still pending, in some cases a replacement may have already been seen on an OAI follow-up x-ray (ExxRPSN=‘1’). For most purposes these can be treated as confirmed joint replacements though details such as the date and reason for the replacement will not yet be available.
- ‘2: There are no follow-up x-rays for this knee (or hip):’ this value is assigned if there are no follow-up x-rays available for the joint.

To select all participants with follow-up replacements seen on x-ray use ExxRPSN=1 for both sides for the joint of interest.

In general, when the only information about a knee or hip replacement is self-reported and not adjudicated (ExxRPCF=1 or 2 and ExxRPSN=0 or 2), these joints should be treated as having had a replacement in analyses. Similarly, if analyses require participants who have not had a replacement, self-reported replacements should be removed.

Description of other replacement variables:

Date of follow-up replacement. The date of a follow-up replacement is provided in the variable ExxDATE. The date of replacement obtained from medical records is provided when available. If the date of replacement is not available from the medical records, the date reported by the participant is provided if available. The flag variable ExxFLDT indicates if the date of replacement is self-reported or from the medical records. If the date is self-reported, the date of replacement may change in a future release after the medical records have been received and the date of replacement adjudicated.

Closest OAI contact prior to and after replacement. Several variables in the Outcomes dataset give information on the closest contact prior to or closest visit after a replacement. If any interview data, clinical data and/or images were collected, this is defined as a contact. Please see the variable VISITYP in the Subject Characteristics/Risk Factors Outcomes_descrip 3/1/2016

dataset or AllClinical dataset to determine the type of visit a participant had. Note the 60- and 84-month contacts are telephone interviews only.

Table 2 shows which variables to use for determining OAI contacts before and after replacements. In general the replacement in question will be between the contacts indicated by the variables V99ExxVSPR and V99ExxVSAF (see Tables 3 and 4 for the values and labels that these variables can take).

The actual number of days between the enrollment visit and the date of replacement is given in the variables V99ERKDAY (R knee replacements), V99ELKDAY (L knee replacements), V99ERHDAY (R hip replacements), V99ELHDAY (L hip replacements). These variables are particularly useful when the closest contact after the replacement has missing data, which most likely occurs when the participant did not return to the clinic for a visit, and the report of the replacement was by phone interview.

Table 2. Pairs of variables that can be used for determining OAI contacts before and after the date of knee and hip replacement				
	RIGHT		LEFT	
	Closest Contact prior to replacement	Closest Contact after replacement	Closest Contact prior replacement	Closest Contact after replacement
Knee Replacement	V99ERKVSPR	V99ERKVSAF	V99ELKVSPR	V99ELKVSAF
Hip Replacement	V99ERHVSPR	V99ERHVSAF	V99ELHVSPR	V99ELHVSAF

Some participants will have missing data at their closest contact after replacement. To determine which participants have withdrawn from the study or are lost to follow-up, please consult the REASON variable in the Subject Characteristics/Risk Factors Dataset.

Description of death variables:

EDDCF is the key variable for identifying deaths. For participants for whom a death was reported, this variable gives information on where participants are in the adjudication process and has three possible values:

- ‘0: Death reported, no documentation available:’ this value is assigned when a death is reported and documentation confirming the death will not be obtained.
- ‘1: Death pending adjudication:’ this value is assigned if a death has been reported, but adjudication process has not yet been completed. In these cases, documentation from the clinical center is pending that confirms the death. This value may change in subsequent releases of the Outcomes dataset as documentation is received for these participants.
- ‘2: Death adjudicated, confirmed:’ this value is assigned if a death has been confirmed through the adjudication process.

The date of death is provided in the variable EDDDATE. The date of death from documentation (e.g., death certificate or obituary) is provided when available. If this date

is not available, the reported date of death is provided if available. The flag variable EDDFLDT indicates if the date of death is reported or obtained from documentation. If the date is reported, the date of death may change in a future release after documentation has been received and the date of death adjudicated.

Description of x-ray outcome variables:

As of version 2, x-ray outcome variables have been added to the Outcomes dataset. All the x-ray outcome related variables in this dataset (variable names prefixed V99ERX... or V99ELX..., for right and left knee respectively) were modified in version 6 to emphasize that two different definitions of incident radiographic OA (ROA) were used. This includes a weak definition which was development of KLG ≥ 2 in a knee that was KLG 0 or 1 at baseline, regardless of whether a knee had joint space narrowing or not. These knees could develop ROA simply due to enlarging osteophytes, in a knee with normal joint space. The stronger definition of ROA required the development of KLG ≥ 2 in a knee that was KLG 0 or 1 at baseline, along with presence of JSN (OARSI JSN grade ≥ 1 at or before the visit of incidence).

Variables V99ERXIOA indicates the first visit at which the weak definition of incident ROA was met and V99ERXKNOA indicates the last visit at which the weak definition of ROA was not met for right knees. Variables V99ELXIOA and V99ELXKNOA are the matching variables for left knees. For the stronger definition of incident ROA, requiring KLG ≥ 2 AND JSN ≥ 1 , there are 4 similar variables: V99ERXIOAN and V99ERXNOAN (right knee) and V99ELXIOAN and V99ELXNOAN (left knee). Table 6 shows the values and labels for these variables related to incident ROA.

Since follow-up visit x-rays after baseline were read blinded to order, there are some knees where KLG < 2 at the participants first and last x-ray visit, but may be KLG ≥ 2 at an intermediate time point. As of version 7, we have provided 4 variables in the outcomes dataset to enable users to easily identify such knees. There are two variables for the right knee (V99ERKLOA = right knee, last visit with KLG ≥ 2 , V99ERKLOAN = right knee, last visit with KLG ≥ 2 and JSN ≥ 1) and there are two equivalent variables (V99ELKLOA and V99ELKLOAN) for the left knee. If these variables are later than the related “first visit” with OA variables (V99ERXIOA, V99ERXIOAN, V99ELXIOA, V99ELXIOAN), then this indicates that the knee has OA at an intermediate time point, but not at the participant’s last knee x-ray visit. The documentation for the OAI SAS datasets kXR_SQ_BUxx contains further information about this type of situation.

The x-ray outcome variables in the Outcomes dataset are based on data from the OAI Projects 15 and 37 which consisted of semi-quantitative x-ray readings of knees in OAI for which there were follow-up knee radiographs available. Further information about these projects can be found in kXR_SQ_BU_Descrip.pdf . The outcome variables in this dataset are calculated from the raw data from Projects 15 and 37 contained in the OAI SAS datasets kXR_SQ_BUxx. Please see Section 1.3 and Appendix A of that document for further information about the x-ray outcomes variables.

As of version 8, rather than just providing outcome variables for whether a knee had joint space narrowing (JSN) progression over follow-up, we now provide summary variables

for the first visit at which JSN had progressed, which tibio-femoral compartment initially narrowed, and also summary variables for the last visit at which JSN was no worse than at baseline and the last visit at which JSN was narrowed compared to baseline. These variables are listed in Table 6 and further descriptions of them are in Section 1.3 and Appendix A of the documentation for the kXR_SQ_BUxx datasets kXR_SQ_BU_descrip.pdf. These variables exist for 2 definitions of JSN Progression: (i) when the JSN grade is (whole grade or more) worse than at baseline, and (ii) when the JSN grade is (partial grade or more) worse than at baseline.

Please note that the x-ray readings which the radiographic outcome variables were based were done blinded to the order of follow-up visits. Therefore, there are likely to be some participants where the outcome of incident radiographic knee OA is met at an early follow-up visit, but for whom the knee has a Kellgren and Lawrence grade < 2 at their final visit.

Description of other variables:

As of version 7, the variable V99RNTCNT was added to the Outcomes dataset. This variable reports the most recent OAI contact the participant had. If any interview data, clinical data and/or images were collected, this is defined as a contact.

Data structure/contents:

This dataset contains one record per participant. Surgical revisions of replacements are not included in this dataset. The variable uniquely identifying a record is ID, and the datasets are sorted by ID, which can be used to merge/join to data in other datasets.

A full list and description of all the variables contained in this dataset can be found below and in the contents.pdf.

Condition of data:

- **Known data errors:** problems/cautions for use are listed by variable in the “Release Comments.”
- **Dataset strengths/weaknesses:** None at this time.
- **Data changes:**
 - As of version 2, one participant (9332345) had the left knee visit reported at variable, V99ELKVSRP, change from 48-month to 60-month because although the replacement occurred after the 48-month visit, it was inadvertently coded as being reported at the 48-month visit.
 - As of version 6, the following variables were added to the Outcomes dataset: V99ERXIOA, V99ELXIOA, V99ERXIOAN, V99ELXIOAN, V99ERXNOA, V99ELXNOA, V99ERXNOAN, V99ELXNOAN; and the following variables were removed from the Outcomes dataset: V99ERXKL2, V99ELXKL2, V99ERXKL2N, V99ELXKL2N.
 - As of version 7, the following variables were added to the Outcomes dataset: V99RNTCNT, V99ERKLOA, V99ERKLOAN, V99ELKLOA, V99ELKLOAN.
 - As of version 8, the following variables were added to the Outcomes dataset: V99ERJSFW, V99ERNJSLW, V99ERNJSLW, V99ERJSTFW,

V99ERJSFP, V99ERJSLP, V99ERNJSLP, V99ERJSTFP, V99ELJSFW, V99ELJSLW, V99ELNJSLW, V99ELJSTFW, V99ELJSFP, V99ELJSLP, V99ELNJSLP, V99ELJSTFP; and the following variables were removed from the Outcomes dataset: V99ERXJSNM, V99ERXJSNL, V99ELXJSNM, and V99ELXJSNL.

General strategies for use:

When using with other datasets, merge/join by ID.

Analysts are encouraged to always output and view SAS variable labels in their entirety to ensure important information about the variables is not lost. The maximum SAS label length is 160 characters.

Table 3. Knee Replacements

Variable Name	Variable Description	Values + Value Labels
V99ERKVSRP	OAI visit follow-up knee replacement (right knee) self-reported at	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month
V99ELKVSRP	OAI visit follow-up knee replacement (left knee) self-reported at	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month
V99ERKRPCF	Follow-up knee replacement (right knee) adjudication/confirmation status.	0: Replacement adjudicated and failed to confirm; 1: Self-reported replacement, 2: Replacement pending adjudication, 3: Replacement adjudicated, confirmed
V99ELKRPCF	Follow-up knee replacement (left knee) adjudication/confirmation status	0: Replacement adjudicated and failed to confirm; 1: Self-reported replacement, 2: Replacement pending adjudication, 3: Replacement adjudicated, confirmed
V99ERKRPSN	Knee replacement (right knee) seen on follow-up OAI x-ray	0: No replacement is seen on any available follow-up x-ray; 1: Yes, replacement is seen on one or more available follow-up x-rays; 2: There are no follow-up x-rays of this knee (or hip)
V99ELKRPSN	Knee replacement (left knee) seen on follow-up OAI x-ray	0: No replacement is seen on any available follow-up x-ray; 1: Yes, replacement is seen on one or more available follow-up x-rays; 2: There are no follow-up x-rays of this knee (or hip)
V99ERKDATE	Date of follow-up knee replacement (right knee)	Date
V99ELKDATE	Date of follow-up knee replacement (left knee)	Date
V99ERKFLDT	Date flag, date of follow-up knee replacement (right knee) from self report or adjudicated from medical records	0: Self-report, 1: Medical records
V99ELKFLDT	Date flag, date of follow-up knee replacement (left knee) from self report or adjudicated from medical records	0: Self-report, 1: Medical records
V99ERKTLPR	Total or partial follow-up knee replacement (right knee)	1: Total , 2: Partial
V99ELKTLPR	Total or partial follow-up knee replacement (left knee)	1: Total , 2: Partial
V99ERKTPPR	Type of partial follow-up knee replacement (right knee)	1: Partial lateral, 2: Partial medial, 3: Partial patello-femoral, 4: Unable to determine
V99ELKTPPR	Type of partial follow-up knee replacement (left knee)	1: Partial lateral, 2: Partial medial, 3: Partial patello-femoral, 4: Unable to determine
V99ERKPODX	Primary pre-operative diagnosis (right knee)	1: Osteoarthritis or degenerative arthritis, 2: Rheumatoid arthritis, 3: Acute fracture or acute trauma, 4: Post-traumatic arthritis (old trauma), 5: Osteonecrosis, 6: Revision/salvage of previous replacement, 7: Unable to determine from doc provided, 8: Other

Table 3. Knee Replacements (continued)

Variable Name	Variable Description	Values + Value Labels
V99ELKPODX	Primary pre-operative diagnosis (left knee)	1: Osteoarthritis or degenerative arthritis, 2: Rheumatoid arthritis, 3: Acute fracture or acute trauma, 4: Post-traumatic arthritis (old trauma), 5: Osteonecrosis, 6: Revision/salvage of previous replacement, 7: Unable to determine from doc provided, 8: Other
V99ERKBLRP	Knee replacement (right knee) seen on baseline OAI x-ray	0: No, 1: Yes
V99ELKBLRP	Knee replacement (left knee) seen on baseline OAI x-ray	0: No, 1: Yes
V99ERKDAY5	Days between enrollment visit and follow-up knee replacement (right knee)	Continuous number
V99ELKDAY5	Days between enrollment visit and follow-up knee replacement (left knee)	Continuous number
V99ERKVSPR	Closest OAI contact prior to follow-up knee replacement (right knee)	0: Baseline, 1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month
V99ELKVSPR	Closest OAI contact prior to follow-up knee replacement (left knee)	0: Baseline, 1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month
V99ERKVSAF	Closest OAI contact after follow-up knee replacement (right knee)	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month, 99: No post-replacement follow-up x-ray
V99ELKVSAF	Closest OAI contact after follow-up knee replacement (left knee)	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month, 99: No post-replacement follow-up x-ray
V99ERKXRPR	Closest OAI visit with knee x-ray prior to follow-up knee replacement (right knee)	0: Baseline, 1: 12-month, 3: 24-month, 5: 36-month, 6: 48-month, 8: 72-month, 10: 96-month
V99ELKXRPR	Closest OAI visit with knee x-ray prior to follow-up knee replacement (left knee)	0: Baseline, 1: 12-month, 3: 24-month, 5: 36-month, 6: 48-month, 8: 72-month, 10: 96-month
V99ERKXRAF	Closest OAI visit with knee x-ray after follow-up knee replacement (right knee)	1: 12-month, 3: 24-month, 5: 36-month, 6: 48-month, 8: 72-month, 10: 96-month, 99: No post-replacement follow-up x-ray
V99ELKXRAF	Closest OAI visit with knee x-ray after follow-up knee replacement (left knee)	1: 12-month, 3: 24-month, 5: 36-month, 6: 48-month, 8: 72-month, 10: 96-month, 99: No post-replacement follow-up x-ray

Table 4. Hip Replacements		
Variable Name	Variable Description	Values + Value Labels
V99ERHVS RP	OAI visit follow-up hip replacement self-reported at (right hip)	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month
V99ELHVS RP	OAI visit follow-up hip replacement self-reported at (left hip)	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month
V99ERHRPCF	Follow-up hip replacement (right hip) adjudication/confirmation status	0: Replacement adjudicated and failed to confirm; 1: Self-reported replacement, 2: Replacement pending adjudication, 3: Replacement adjudicated, confirmed
V99ELHRPCF	Follow-up hip replacement (left hip) adjudication/confirmation status	0: Replacement adjudicated and failed to confirm; 1: Self-reported replacement, 2: Replacement pending adjudication, 3: Replacement adjudicated, confirmed
V99ERHRPSN	Hip replacement (right hip) seen on follow-up OAI x-ray	0: No replacement is seen on any available follow-up x-ray; 1: Yes, replacement is seen on one or more available follow-up x-rays; 2: There are no follow-up x-rays of this knee (or hip)
V99ELHRPSN	Hip replacement (left hip) seen on follow-up OAI x-ray	0: No replacement is seen on any available follow-up x-ray; 1: Yes, replacement is seen on one or more available follow-up x-rays; 2: There are no follow-up x-rays of this knee (or hip)
V99ERHDATE	Date of follow-up hip replacement (right hip)	Date
V99ELHDATE	Date of follow-up hip replacement (left hip)	Date
V99ERHFLDT	Date flag, date of follow-up hip replacement (right hip) from self report or adjudicated from medical records	0: Self-report, 1: Medical records
V99ELHFLDT	Date flag, date of follow-up hip replacement (left hip) from self report or adjudicated from medical records	0: Self-report, 1: Medical records
V99ERHPODX	Primary pre-operative diagnosis (right hip)	1: Osteoarthritis or degenerative arthritis, 2: Rheumatoid arthritis, 3: Acute fracture or acute trauma, 4: Post-traumatic arthritis (old trauma), 5: Osteonecrosis, 6: Revision/salvage of previous replacement, 7: Unable to determine from doc provided, 8: Other
V99ELHPODX	Primary pre-operative diagnosis (left hip)	1: Osteoarthritis or degenerative arthritis, 2: Rheumatoid arthritis, 3: Acute fracture or acute trauma, 4: Post-traumatic arthritis (old trauma), 5: Osteonecrosis, 6: Revision/salvage of previous replacement, 7: Unable to determine from doc provided, 8: Other
V99ERHBLRP	Hip replacement (right hip) seen on baseline OAI x-ray	0: No, 1: Yes
V99ELHBLRP	Hip replacement (left hip) seen on baseline OAI x-ray	0: No, 1: Yes

Table 4. Hip Replacements (continued)		
Variable Name	Variable Description	Values + Value Labels
V99ERHDDAYS	Days between enrollment visit and follow-up hip replacement (right hip)	Continuous number
V99ELHDDAYS	Days between enrollment visit and follow-up hip replacement (left hip)	Continuous number
V99ERHVSPPR	Closest OAI contact prior to follow-up hip replacement (right hip)	0: Baseline, 1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month
V99ELHVSPPR	Closest OAI contact prior to follow-up hip replacement (left hip)	0: Baseline, 1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month
V99ERHVSAPF	Closest OAI contact after follow-up hip replacement (right hip)	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month, 99: No post-replacement follow-up x-ray
V99ELHVSAPF	Closest OAI contact after follow-up hip replacement (left hip)	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month, 99: No post-replacement follow-up x-ray
V99ERHXRP	Closest OAI visit with hip x-ray prior to follow-up hip replacement (right hip)	0: Baseline, 1: 12-month, 6: 48-month, 10: 96-month
V99ELHXRP	Closest OAI visit with hip x-ray prior to follow-up hip replacement (left hip)	0: Baseline, 1: 12-month, 6: 48-month, 10: 96-month
V99ERHXRAF	Closest OAI visit with hip x-ray after follow-up hip replacement (right hip)	1: 12-month, 6: 48-month, 10: 96-month, 99: No post-replacement follow-up x-ray
V99ELHXRAF	Closest OAI visit with hip x-ray after follow-up hip replacement (left hip)	1: 12-month, 6: 48-month, 10: 96-month, 99: No post-replacement follow-up x-ray

Table 5. Deaths

Variable Name	Variable Description	Values + Value Labels
V99EDDCF	Death adjudication/confirmation status	0: Death reported, no documentation available 1: Death pending adjudication 2: Death adjudicated, confirmed
V99EDDDATE	Date of death	Date
V99EDDFLDT	Date flag, date of death from documentation or from reported date of death	0: Reported date 1: Confirmed date from documentation
V99EDDVSPR	Closest OAI visit prior to death	1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month

Table 6. X-ray Outcomes

Variable Name	Variable Description	Values + Value Labels
V99ERXIOA	Summary incident TF ROA KL >= 2 (right knee)	1: KL>=2 at BL 2: KL0-1 at BL, follow-up x-rays but no incident KL >= 2 3: KL0-1 at BL, incident KL2 at 12 mo 4: KL0-1 at BL, incident KL2 at 24 mo 5: KL0-1 at BL, incident KL2 at 36 mo 6: KL0-1 at BL, incident KL2 at 48 mo 7: KL0-1 at BL, incident KL2 at 72 mo 8: KL0-1 at BL, incident KL2 at 96 mo
V99ELXIOA	Summary incident TF ROA KL >= 2 (left knee)	1: KL>=2 at BL 2: KL0-1 at BL, follow-up x-rays but no incident KL >= 2 3: KL0-1 at BL, incident KL2 at 12 mo 4: KL0-1 at BL, incident KL2 at 24 mo 5: KL0-1 at BL, incident KL2 at 36 mo 6: KL0-1 at BL, incident KL2 at 48 mo 7: KL0-1 at BL, incident KL2 at 72 mo 8: KL0-1 at BL, incident KL2 at 96 mo
V99ERXIOAN	Summary incident TF ROA KL >= 2 with JSN (right knee)	1: KL>=2 with JSN at BL 2: KL0-1 at BL with JSN, follow-up x-rays but no incident KL>=2 with JSN 3: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 12 mo 4: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 24 mo 5: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 36 mo 6: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 48 mo 7: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 72 mo 8: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 96 mo

Table 6. X-ray Outcomes (continued)

Variable Name	Variable Description	Values + Value Labels
V99ELXIOAN	Summary incident TF ROA KL >= 2 with JSN (left knee)	1: KL>=2 with JSN at BL 2: KL0-1 at BL with JSN, follow-up x-rays but no incident KL>=2 with JSN 3: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 12 mo 4: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 24 mo 5: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 36 mo 6: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 48 mo 7: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 72 mo 8: KL0-1 or KL2 with no JSN at BL, incident KL>=2 w/JSN at 96 mo
V99ERXNOA	Last visit KL < 2 (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ELXNOA	Last visit KL < 2 (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ERXNOAN	Last visit KL < 2 or KL=2 with no JSN (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ELXNOAN	Last visit KL < 2 or KL=2 with no JSN (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month

Table 6. X-ray Outcomes (continued)

Variable Name	Variable Description	Values + Value Labels
V99ERKLOA	Last visit with OA (KLG>=2) (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ELKLOA	Last visit with OA (KLG>=2) (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ERKLOAN	Last visit with OA (KLG>=2 and JSN>=1) (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ELKLOAN	Last visit with OA (KLG>=2 and JSN>=1) (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month

Table 6. X-ray Outcomes (continued)

Variable Name	Variable Description	Values + Value Labels
V99ERJSFW	First visit with JSN progression (whole grade or more) (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ELJSFW	First visit with JSN progression (whole grade or more) (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ERJSLW	Last visit with JSN progression (whole grade or more) (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ELJSLW	Last visit with JSN progression (whole grade or more) (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ERNJSLW	Last visit with no JSN progression (whole grade or more) (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month

Table 6. X-ray Outcomes (continued)

Variable Name	Variable Description	Values + Value Labels
V99ELNJSLW	Last visit with no JSN progression (whole grade or more) (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ERJSTFW	TF compartment with initial JSN progression (whole grade or more) (right knee)	0: None 1: Lateral 2: Medial 3: Medial and Lateral
V99ELJSTFW	TF compartment with initial JSN progression (whole grade or more) (left knee)	0: None 1: Lateral 2: Medial 3: Medial and Lateral
V99ERJSFP	First visit with JSN progression (partial grade or more) (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ELJSFP	First visit with JSN progression (partial grade or more) (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ERJSLP	Last visit with JSN progression (partial grade or more) (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month

Table 6. X-ray Outcomes (continued)

Variable Name	Variable Description	Values + Value Labels
V99ELJSLP	Last visit with JSN progression (partial grade or more) (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ERNJSLP	Last visit with no JSN progression (partial grade or more) (right knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ELNJSLP	Last visit with no JSN progression (partial grade or more) (left knee)	0: Baseline 1: 12-month 3: 24-month 5: 36-month 6: 48-month 8: 72-month 10: 96-month
V99ERJSTFP	TF compartment with initial JSN progression (partial grade or more) (right knee)	0: None 1: Lateral 2: Medial 3: Medial and Lateral
V99ELJSTFP	TF compartment with initial JSN progression (partial grade or more) (left knee)	0: None 1: Lateral 2: Medial 3: Medial and Lateral
V99EXLVSQD	Last visit with central reading follow-up knee x-ray SQ (KL or JSN) data	0: No follow-up knee x-ray or SQ (KL or JSN) data 1: Last SQ data at 12 mo visit 2: Last SQ data at 24 mo visit 3: Last SQ data at 36 mo visit 4: Last SQ data at 48 mo visit 5: Last SQ data at 72 mo visit 6: Last SQ data at 96 mo visit

Table 7.

Miscellaneous		
Variable Name	Variable Description	Values + Value Labels
V99RNTCNT	Most recent OAI contact	0: Baseline, 1: 12-month, 2: 18-month, 3: 24-month, 4: 30-month, 5: 36-month, 6: 48-month, 7: 60-month, 8: 72-month, 9: 84-month, 10: 96-month, 11: 108-month